

IECEX/ATEX Technical Certification Document — Gas Leak Detector (GLD) V2

Prepared in direct reference to the *IECEX/ATEX Certification Information Requirements* issued by the certification body (ExCB) — Section **2, Technical Documentation**, Items 1–4: product description, product name/model/specification list, functional description and technical parameters, and product photographs. This document is being developed in stages; remaining items will follow in subsequent revisions.

CERTIFICATION SUBJECT	MANUFACTURER	TECHNICAL PARTNER	REFERENCE CHECKLIST
Node Sensor (GLD) — V2	PT LAPI Ganesha Utama	Institute of Technology Bandung	IECEX/ATEX Certification Info. Requirements

• About this document

This document is a working technical file prepared to satisfy Section “**2. Technical Documentation**” of the official *IECEX/ATEX Certification Information Requirements* checklist issued by the certification body (ExCB). The table of contents on the left follows the complete structure of the original checklist (Sections 1–3); sections not yet completed are marked with a pending status and will follow in subsequent revisions.

Current certification scope — Node Sensor (GLD) only. The Cluster Head and LoRa Gateway units are assumed to operate exclusively in non-hazardous (safe) areas and therefore fall **outside** the scope of this IECEX/ATEX certification — this is a project working assumption, not the outcome of a formal area classification study by the end client. Specification tables for the Cluster Head and Gateway are included in Sections 2.2–2.3 for **system context only** (the three device types share significant architecture and components), not as certification subjects.

Original excerpt, Section 2, Items 1–4 — source: IECEX ATEX Certification Information Requirements (ExCB): “1) Detailed product description; 2) Product name, model, and specification list; 3) Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.); 4) Clear product photos (overall and key components).”

1 Basic Information (Application and Organization)

Covers the application form, business license/company registration, organizational chart and contact information, manufacturing facility address and profile, and (where applicable) ISO 9001 certification.

- ☐ **Not yet prepared in this revision.** To be completed in a subsequent revision.

2 Technical Documentation

Nine items per the original checklist (1–9, with Item 6 comprising sub-items a–i). This revision addresses Items 1–4; Items 5–9 will follow.

2.1 · Detailed Product Description

The Gas Leak Detector (GLD) Node Sensor is an IoT-based, multi-sensor gas leak detection device designed for the early detection of flammable and process gases in oil & gas refinery environments (process units, tank farms, pipe racks, and storage/loading-unloading areas). It is one of three device types within the broader GLD V2 system — Node Sensor (GLD), Cluster Head, and LoRa Gateway — however **only the Node Sensor is the subject of the current IECEx/ATEX certification** (see scope note above).

Functionally, the Node Sensor integrates eight channels of metal-oxide semiconductor gas sensors (MQ series), an edge-AI microcontroller/processor (ESP32-S3), and a LoRa radio module (star topology to the nearest Cluster Head) within a single fixed-point unit installed at locations with gas-leak risk. An on-device AI gas-classification model (CNN Dual-Branch architecture) runs directly on the unit — *confirmed by the engineering team; written verification within firmware documentation has not yet been completed* — so that detection decisions do not depend on a continuous connection to a central server. When gas concentration exceeds a defined threshold, the unit triggers a local alarm (an integrated visual/audible alarm module) and simultaneously transmits an alarm notification over the LoRa network to the Cluster Head, which forwards it to the operator dashboard.

The enclosure is designed for hazardous-area deployment at refinery sites, using metal materials (aluminum alloy and stainless steel — no plastic or PVC) and mounted via an L-bracket to existing structures without drilling or welding. **Important:** this design-intent statement does not constitute a claim that the enclosure has passed testing or has been Ex-certified — the explosion-protection scheme, gas group, temperature class, and target installation zone will be addressed in Section 2.5 (to follow).

The current production power configuration is continuous 24 VDC, supplied via an AC/DC adapter connected to the site electrical supply. A portable battery power path (Li-ion 18650) remains under development (R&D) and has not become a deployed production configuration.

2.2 · Product Name, Model, and Specification List

Product name Gas Leak Detector (GLD) — Node Sensor	Model / version GLD V2 (Version 2)	Related system devices LoRa Cluster Head, LoRa Gateway (outside certification scope)
Manufacturer PT LAPI Ganesha Utama	Technical development partner Institute of Technology Bandung — IoT Laboratory & Physics Laboratory	End client / program owner PT Pertamina Patra Niaga (initial deployment site: Refinery Unit IV, Cilacap)

The table below summarizes indicative specifications for all three device types within the GLD V2 system, provided for system-architecture context. **The certification subject is the “Node Sensor (GLD)” column only.**

PARAMETER	NODE SENSOR (GLD)	CLUSTER HEAD	LORA GATEWAY
Primary function	Acquisition of 8-channel gas sensor data and LoRa transmission	Aggregates data from multiple Node Sensors and forwards to the Gateway	Bridges the field LoRa network to the server
Model/version	V2	V2	V2
Microcontroller	ESP32-S3-WROOM-1U	ESP32-S3-WROOM-1U	ESP32-S3-WROOM-1U
LoRa radio module	E22-900MM22S	E22-900MM22S	E22-900MM22S
Power input	24 VDC, ≈0.33 A	5 VDC (battery + solar)	5 VDC, AC/DC adapter
Max. power consumption	7.995 W	0.73 W	0.73 W
Dimensions (L×W×H)	200 × 90 × 290 mm	80 × 80 × 210 mm	80 × 80 × 210 mm
Enclosure material	Aluminum alloy + stainless steel	Aluminum alloy + stainless steel	Aluminum alloy + stainless steel
IP rating	Pending confirmation	IP66/67	Likely equivalent to the Cluster Head; not yet confirmed separately
Certification subject (ATEX/IECEx)?	Yes — the subject of this document	No (assumed safe area)	No (assumed safe area)

Source: internal technical specification documentation (Revision 27 August 2026) and EMC parameter measurement data (Institute of Technology Bandung, Physics Laboratory).

2.3 · Functional Description and Technical Parameters (Electrical, Mechanical, etc.)

Functional workflow (normal operating mode): sense → process → transmit. Each of the eight gas sensors continuously samples ambient conditions → data is normalized and classified by the on-device AI model (ESP32-S3) → the result is transmitted via LoRa to the Cluster Head (star topology) at a configurable interval (default 10 seconds), or immediately (event-driven) when an alarm condition is detected. Gas alarms are triggered through two parallel channels: a local visual/audible alarm module on the unit itself, and a push notification transmitted over the LoRa network to the dashboard — the alarm-push pathway has been successfully tested on a campus mesh network (field validation at a production refinery installation is still pending).

2.3.A · ELECTRICAL PARAMETERS — NODE SENSOR (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Main power input	24 VDC	Final — production configuration	Continuous power supply from a site AC/DC adapter.
AC/DC adapter input	220 VAC, 50 Hz	Final	Compliant with the Indonesian national electrical grid standard (PLN).
AC/DC adapter output	24 VDC	Final	
Internal operating voltage	5 VDC & 3.3 VDC	Final	5 VDC for the MQ sensor/heater circuitry; 3.3 VDC for ESP32-S3 logic. Per-rail current has not yet been documented separately.
Maximum input current	≈0.33 A @ 24 VDC	Calculated	Calculated from measured maximum power consumption (7.995 W) divided by 24 VDC.
Maximum power consumption	7.995 W @ 24 VDC	Measured — production configuration	Applies to the continuous-power configuration. The battery (R&D) configuration is recorded separately at 5.75 W — a different operating mode, not a data conflict.
Backup battery path (R&D, not yet in production)	Li-ion 18650 cells, 7 cells in parallel, 4.2 V/cell, ≈28,000 mAh total	Development pathway	Not yet a deployed production configuration. Cell/BMS safety certification (e.g., UN 38.3, IEC 62133) has not yet been obtained.
Electrical protection (fuse, reverse polarity,	—	Pending confirmation	Scope of electrical protection has not yet

overvoltage,
overcurrent)

been
defined/documented.

2.3.B · SENSOR AND COMMUNICATION PARAMETERS — NODE SENSOR (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Gas sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135 (8 channels)	Final	Metal-oxide semiconductor sensors; the sensing element is directly exposed to ambient air (not enclosed).
AI gas-classification model	CNN Dual-Branch — 4 classes: LPG, CO ₂ , Clean Air, H ₂	Confirmed by the engineering team; written verification in firmware documentation pending	On-chip accuracy 99.20% (int8, ESP32-S3, model size 9.14 KB). Does not yet cover Benzene, CO, or H ₂ S (an additional client requirement, still open).
Environmental sensor (temperature/humidity)	Not installed on production units	Laboratory test rig only	Used only on the laboratory test rig, not on field units.
Processing unit	ESP32-S3-WROOM-1U	Final	Certified under FCC (2AC7Z-ESPS3WROOM1U), TELEC, and CE (per manufacturer data, Espressif) — these are RF/EMC certifications, not an “Ex component” certification.
Communication module	LoRa, E22-900MM22S module	Final	Certified under CE, FCC, and RoHS (per manufacturer data, Ebyte) — RF/EMC certifications, not an “Ex component” certification.

Operating frequency	920 MHz	Final	Star topology to the Cluster Head; within the regional 920–923 MHz ISM band (Indonesia).
Transmit power (firmware configuration)	17 dBm	Final	The radio module supports up to 22 dBm — 17 dBm is an operational configuration, not the module's maximum limit.
Bandwidth / spreading factor	125 kHz / SF7	Final	Source: EMC parameter table.
Antenna	External, omnidirectional, SMA male connector, 3 dBi gain	Final	On some units, the 2.4 GHz Wi-Fi antenna remains inside the enclosure and must be relocated externally for optimal channel configuration.
Data transmission interval	Configurable, default 10 seconds	Final	Alarm events are transmitted immediately, independent of the periodic interval.
Other interfaces	SPI, LoRa	Final	External ports/connectors: USB, sensor, power, fan, antenna, alarm buzzer.

2.3.C · MECHANICAL PARAMETERS — NODE SENSOR (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Enclosure material	Aluminum alloy + stainless steel	Specific grade pending	PVC is not used in any housing or bracket component. The specific grade (aluminum series, stainless steel 304/316L) is still under determination.

Dimensions (L × W × H)	200 × 90 × 290 mm	Final	Consistent between the technical specification documentation and the EMC parameter table.
Total weight	—	Pending confirmation	Not yet weighed/documented.
Mounting method	L-bracket, following the design already installed at the refinery	Final	Mounted to existing structures without drilling or welding.
Ingress protection (IP rating)	—	Pending confirmation	Not yet tested/determined for the Node Sensor (the Cluster Head is already rated IP66/67).
Cable entry (gland)	—	Pending confirmation	Cable gland specification not yet determined.
Antenna mounting	External, SMA male connector	Final	
Operating temperature	—	Pending confirmation	Ambient operating temperature range not yet determined — a key parameter for temperature class (T1–T6) determination in Section 2.5.
Operating humidity	—	Pending confirmation	

Source: internal technical specification documentation, Sections 1.1–1.3 (Node Sensor), cross-referenced with EMC parameter data and component certification verification (ESP32-S3-WROOM-1U, E22-900MM22S).

 **Rows marked “Pending confirmation” above do not reflect a documentation oversight — this is an honest status indicator.** These fields are intentionally left blank because no official data yet exists (not yet measured, tested, or decided). They must not be filled with estimates in future revisions without a clear supporting data source.

2.4 · Product Photographs — Overall and Key Components

The seven photographs below were taken directly from the Node Sensor (GLD) V2 prototype unit (no capture-date metadata is available in the source files). The first three photographs show the fully assembled unit from different angles; the remaining four show key components in close-up.



Complete unit — front view

The fully assembled Node Sensor (GLD): the main enclosure (blue) with sensor mesh cover, external LoRa antenna, and a visual alarm module (cylindrical housing, red lens) mounted on the right side of the enclosure.



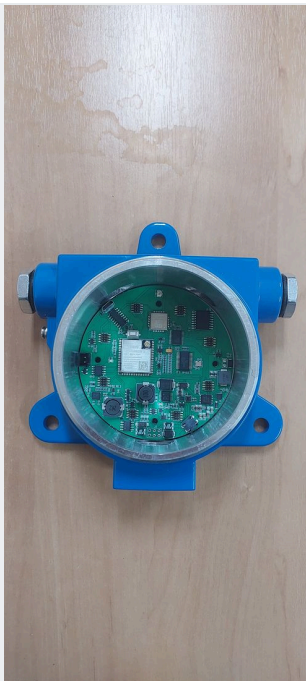
Complete unit — alternate angle

An alternate view of the assembled unit, showing the position of the antenna, alarm module, and the cable-gland side of the enclosure.



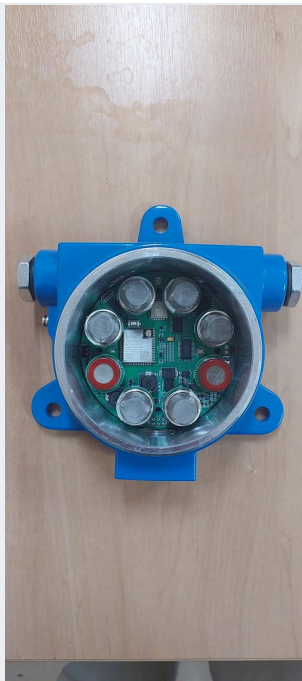
Complete unit — front view (without antenna and alarm module)

The main enclosure with the sensor mesh cover installed, shown in a flat front view illustrating the enclosure shape and its four mounting bolt holes.



Key component — PCB inside enclosure (before sensor modules)

The main PCB installed inside the enclosure prior to gas sensor module installation. The ESP32-S3-WROOM-1U microcontroller, LoRa radio module, and power circuitry are visible.



Key component — PCB with eight gas sensor modules

The same PCB with eight channels of MQ-series gas sensors (MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135) installed around the central microcontroller.



Key component — visual alarm module (close-up)

The visual alarm (beacon/strobe) module, constructed of metal with a red lens, connected to the main enclosure via a dedicated cable gland.



Key component — sensor mesh cover (close-up)

The metal mesh cover that protects the gas sensing elements while serving as a diffusion path for gas to reach the sensors beneath.

- i **Photographic completeness — status as-is.** The seven photographs above cover the complete unit and its key components (PCB, sensor modules, alarm module, mesh cover), substantively satisfying checklist Item 2.4. Still outstanding: (a) formally labeled photographs of each face (front/back/left/right/top/bottom) with a scale reference, as is customary in ExCB submission packages; (b) separate photographs of individual components such as the battery, gaskets/seals, terminals, and cable glands; (c) photographs of the Cluster Head and Gateway (currently outside certification scope; see scope note).

2.5 · Description of Intended Use and Installation Environment (Gas Group, Temperature Class, Area Classification)

- **Not yet prepared in this revision.** The classification scheme (gas group, temperature class, installation zone) will be finalized based on further technical assessment and confirmed together with the certification body (ExCB / notified body).

2.6 · Design and Manufacturing Information (Items a–i: Technical Drawings, Bill of Materials, Material Datasheets, Manufacturing Process, Calculations, Draft Manual, Nameplate, Ex Component Certificates)

- **Not yet prepared in this revision.** To be completed in a subsequent revision, covering technical drawings, the bill of materials, material datasheets, a manufacturing process description, explosion-protection calculations, a draft operating/installation manual, nameplate information, and Ex component certificates.

3 Sample Information

Model, serial number, and sample status (whether the unit can be powered on and operated), together with any required test fixtures or auxiliary equipment.

- **Not yet prepared in this revision.**

This is a working document, prepared in stages, drafted in direct reference to the *IECEX/ATEX Certification Information Requirements* (original English/Mandarin version issued by the certification body). Data sources: internal technical specification documentation (Revision 27 August 2026), EMC parameter measurement data (Institute of Technology Bandung, Physics Laboratory), and product photography. Fields marked “Pending confirmation” are not yet final and must not be relied upon for procurement or certification purposes without further verification.